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EDUCATION AND TRAINING

2022–present Postdoctoral Fellow, Memorial Sloan Kettering Cancer Center, New York, NY
2020–2023 Resident, Anatomic Pathology, Weill Cornell Medicine / NewYork-Presbyterian Hospital, New York, NY
2012–2020 MD/PhD, New York University Grossman School of Medicine, New York, NY
2005–2009 BS, Biophysics, University of California, Los Angeles, CA — minors in Biomedical Research and in History of Science and Medicine

ACADEMIC APPOINTMENTS

2020–present Trainee, Physician-Scientist Training Program, Department of Pathology and Laboratory Medicine, Weill Cornell Medicine

CERTIFICATION AND LICENSURE

2023–present Medical License, State of New York, No. 324528
2023 Diplomate, American Board of Pathology — Anatomic Pathology

HONORS, AWARDS, AND FELLOWSHIPS

2026–present Tow Foundation Postdoctoral Fellowship, Memorial Sloan Kettering Cancer Center
2026 Finalist, MSK three-minute talk competition, Office of Postdoctoral Affairs
2023–2026 NIH/NCI Institutional Training Grant, T32CA260293, Weill Cornell Medicine, Department of Pathology and Laboratory Medicine (appointed trainee)
2019 Fred S. Mandelbaum Award for excellence in pathology, New York University
2017–2018 NIH/NIAID Institutional Training Grant, T32AI100853, NYU Immunology and Inflammation Training Program
2009 Biomedical Research Summer Scholarship, University of California, Los Angeles

RESEARCH EXPERIENCE

2022–present Postdoctoral Fellow, laboratory of Craig B. Thompson, MD, Memorial Sloan Kettering Cancer Center — metabolite signaling in T cell differentiation and systemic metabolism
2014–2018 Doctoral Research (PhD, Immunology), laboratory of Dan R. Littman, MD/PhD, New York University — *Dissertation: The role of the histone chaperone CAF-1 in silencing Cd4 expression in CD8+ T cells*
2013 Rotation Student, laboratory of Mark R. Philips, MD, New York University — VPS35 in NRAS trafficking and signaling in cancer cells
2010–2012 Research Technician, laboratory of Antoni Ribas, MD/PhD, University of California, Los Angeles — targeted therapy resistance in melanoma; autologous T cell and dendritic cell vaccines for metastatic melanoma
2008–2010 Undergraduate Research Volunteer, laboratory of Kenneth A. Bradley, PhD, University of California, Los Angeles — functional conformations of anthrax toxin receptor 1

PEER-REVIEWED RESEARCH ARTICLES

† First or co-first author.

1. † Ng C, Fung TS, Li D, Kropp KN, Somarribas Patterson LF, Markovitz A, Weinberg DN, Jones O, Kim JY, Zhang G, Koche R, Monetti M, Tang H, He Y, Xu Z, Cai X, Yu Z, Bhagavatula G, Colgan SP, Lin YH, Li Z, Steinert EM, Klebanoff CA, Vardhana SA, Chandel NS, Wu L, Thompson CB. Mitochondrial ATP promotes T cell differentiation through chromatin accessibility. *Cell*. 2026. doi:10.1016/j.cell.2026.08.023. PMID 42727574.

2. Pires S, Yang W, Frigerio S, Louis C, Scott C, Zhou YL, Cardakli E, Tran N, ... **Ng C**, ... Longman RS. Innate lymphoid cells activated by the cytokine TL1A link colitis to emergency granulopoiesis and the recruitment of tumor-promoting neutrophils. *Immunity*. 2026;59(2):372–387.e7. PMID: PMC12981380.
3. Tsai MT, Callaghan R, **Ng C**, Peres-Tintin L, Matthews DB, Stanford N, Ganesh K, Estes MK, Diehl GE. Intestinal epithelial TLR5 signaling promotes barrier-supportive macrophages. *Science Immunology*. 2026;11(115):eadr4057. PMID: PMC13016028.
4. Mesa KR, O'Connor KA, **Ng C**, Salvatore SP, Dolynuk A, Rivera Lomeli M, Littman DR. Niche-specific dermal macrophage loss promotes skin capillary ageing. *Nature*. 2025;648(8092):173–181. PMID: PMC12675291.
5. White Z, Cabrera I, Mei L, Clevenger M, Ochoa-Raya A, Kapustka I, Dominguez JR, Zhou J, Koster KP, Anwar S, Wang Q, **Ng C**, ... Sano T. Gut inflammation promotes microbiota-specific CD4 T cell-mediated neuroinflammation. *Nature*. 2025;643(8071):509–518.
6. Kim DV, Chen A, **Ng C**, Peres-Tintin L, Park SE, Acosta M, ... Diehl GE. E. coli-activated cellular circuit restrains intestinal inflammatory microbiota-specific T cells and protects against colitis. *Cell Reports*. 2025;44(10):116332. PMID: PMC12667601.
7. Nanaware PP, Khan ZN, Clement CC, Shetty M, Mota I, Seltzer ES, ... **Ng C**, Nagayama M, ... Santambrogio L. Role of the afferent lymph as an immunological conduit to analyze tissue antigenic and inflammatory load. *Cell Reports*. 2024;43(6):114311. PMID: PMC11233987.
8. Philips EA, Liu J, Kvalvaag A, Mørch AM, Tocheva AS, **Ng C**, Liang H, Ahearn IM, Pan R, Luo CC, Leithner A, Qin Z, Zhou Y, Garcia-España A, Mor A, Littman DR, Dustin ML, Wang J, Kong XP. Transmembrane domain-driven PD-1 dimers mediate T cell inhibition. *Science Immunology*. 2024;9(93):eade6256. PMID: PMC11166110.
9. Cai X, **Ng CP**, Jones O, Fung TS, Ryu KW, Li D, Thompson CB. Lactate activates the mitochondrial electron transport chain independently of its metabolism. *Molecular Cell*. 2023;83(21):3904–3920.e7. PMID: PMC10752619.
10. Schwörer S, Cimino FV, Ros M, Tsanov KM, **Ng C**, Lowe SW, Carmona-Fontaine C, Thompson CB. Hypoxia potentiates the inflammatory fibroblast phenotype promoted by pancreatic cancer cell-derived cytokines. *Cancer Research*. 2023;83(10):1596–1610. PMID: PMC10658995.
11. Viladomiu M, Khounlotham M, Dogan B, Lima SF, Elsaadi A, Cardakli E, Castellanos JG, **Ng C**, ... Lipkin SM, Longman RS. Agr2-associated ER stress promotes adherent-invasive E. coli dysbiosis and triggers CD103+ dendritic cell IL-23-dependent ileocolitis. *Cell Reports*. 2022;41(7):111637. PMID: PMC9805753.
12. Zegarra Ruiz DF, Kim DV, Norwood K, Saldana-Morales FB, Kim M, **Ng C**, Callaghan R, ... Longman RS, Diehl GE. Microbiota manipulation to increase macrophage IL-10 improves colitis and limits colitis-associated colorectal cancer. *Gut Microbes*. 2022;14(1):2119054. PMID: PMC9450902.
13. Gu W, Wang H, Huang X, Kraicz J, Singh PNP, **Ng C**, Dagdeviren S, ... Shivdasani RA, Zhou Q. SATB2 preserves colon stem cell identity and mediates ileum-colon conversion via enhancer remodeling. *Cell Stem Cell*. 2022;29(1):101–115.e10. PMID: PMC8741647.
14. Sano T, Kageyama T, Fang V, Kedmi R, Martinez CS, Talbot J, Chen A, Cabrera I, Gorshko O, Kurakake R, Yang Y, **Ng C**, Schwab SR, Littman DR. Redundant cytokine requirement for intestinal microbiota-induced Th17 cell differentiation in draining lymph nodes. *Cell Reports*. 2021;36(8):109608. PMID: PMC8845566.
15. Wu L, Hollinshead KER, Hao Y, Au C, Kroehling L, **Ng C**, Lin WY, Li D, Silva HM, Shin J, Lafaille JJ, Possemato R, Pacold ME, Papagiannakopoulos T, Kimmelman AC, Satija R, Littman DR. Niche-selective inhibition of pathogenic Th17 cells by targeting metabolic redundancy. *Cell*. 2020;182(3):641–654.e20. PMID: PMC7556360.
16. † **Ng C**, Aichinger M, Nguyen T, Au C, Najar T, Wu L, Mesa KR, Liao W, Quivy JP, Hubert B, Almouzni G, Zuber J, Littman DR. The histone chaperone CAF-1 cooperates with the DNA methyltransferases to maintain Cd4 silencing in cytotoxic T cells. *Genes & Development*. 2019;33(11–12):669–683. PMID: PMC6546056.
17. Atefi M, Titz B, Tsoi J, Avramis E, Le A, **Ng C**, Lomova A, ... Ribas A, Graeber TG. CRAF R391W is a melanoma driver oncogene. *Scientific Reports*. 2016;6:27454. PMID: PMC4897636.
18. Atefi M, Titz B, Avramis E, **Ng C**, Wong DJL, ... Graeber TG, Ribas A. Combination of pan-RAF and MEK inhibitors in NRAS mutant melanoma. *Molecular Cancer*. 2015;14(1):27. PMID: PMC4320814.
19. Yang Y, Torchinsky MB, Gobert M, Xiong H, Xu M, Linehan JL, Alonzo F, **Ng C**, Chen A, Lin X, Sczesnak A, Liao JJ, Torres VJ, Jenkins MK, Lafaille JJ, Littman DR. Focused specificity of intestinal Th17 cells towards commensal bacterial antigens. *Nature*. 2014;510(7503):152–156. PMID: PMC4128479.

20. Chodon T, Comin-Anduix B, Chmielowski B, Koya RC, Wu Z, Auerbach M, **Ng C**, Avramis E, ... Ribas A. Adoptive transfer of MART-1 T-cell receptor transgenic lymphocytes and dendritic cell vaccination in patients with metastatic melanoma. *Clinical Cancer Research*. 2014;20(9):2457–2465. PMCID: PMC4070853.
21. Shi H, Hong A, Kong X, Koya RC, Song C, Moriceau G, Hugo W, Yu CC, **Ng C**, Chodon T, ... Ribas A, Long GV, Lo RS. A novel AKT1 mutant amplifies an adaptive melanoma response to BRAF inhibition. *Cancer Discovery*. 2014;4(1):69–79. PMCID: PMC3893054.
22. Escuin-Ordinas H, Atefi M, Fu Y, Cass A, **Ng C**, Huang RR, ... Graeber TG, Herschman HR, Ribas A. COX-2 inhibition prevents the appearance of cutaneous squamous cell carcinomas accelerated by BRAF inhibitors. *Molecular Oncology*. 2014;8(2):250–260. PMCID: PMC3943738.
23. Ma C, Cheung AF, Chodon T, Koya RC, Wu Z, **Ng C**, Avramis E, ... Ribas A, Heath JR. Multifunctional T-cell analyses to study response and progression in adoptive cell transfer immunotherapy. *Cancer Discovery*. 2013;3(4):418–429. PMCID: PMC3716460.
24. Hou S, Zhao L, Shen Q, Yu J, **Ng C**, Kong X, ... Ribas A, Lo RS, Tseng HR. Polymer nanofiber-embedded microchips for detection, isolation, and molecular analysis of single circulating melanoma cells. *Angewandte Chemie International Edition*. 2013;52(12):3379–3383. PMCID: PMC3807678.
25. Escuin-Ordinas H, Elliott MW, Atefi M, Lee M, **Ng C**, Wei L, ... Ribas A. PET imaging to non-invasively study immune activation leading to antitumor responses with a 4-1BB agonistic antibody. *Journal for ImmunoTherapy of Cancer*. 2013;1:14. PMCID: PMC4019904.
26. Dahlman KB, Xia J, Hutchinson K, **Ng C**, Hucks D, ... Ribas A, Zhao Z, Pao W. BRAF(L597) mutations in melanoma are associated with sensitivity to MEK inhibitors. *Cancer Discovery*. 2012;2(9):791–797. PMCID: PMC3449158.
27. Shi H, Moriceau G, Kong X, Lee MK, Lee H, Koya RC, **Ng C**, Chodon T, ... Ribas A, Lo RS. Melanoma whole-exome sequencing identifies (V600E)B-RAF amplification-mediated acquired B-RAF inhibitor resistance. *Nature Communications*. 2012;3:724. PMCID: PMC3530385.
28. von Euw E, Atefi M, Attar N, Chu C, Zachariah S, Burgess BL, Mok S, **Ng C**, Wong DJL, ... Ribas A. Antitumor effects of the investigational selective MEK inhibitor TAK733 against cutaneous and uveal melanoma cell lines. *Molecular Cancer*. 2012;11:22. PMCID: PMC3444881.
29. Poulidakos PI, Persaud Y, Janakiraman M, Kong X, **Ng C**, ... Ribas A, Lo RS, Rosen N, Solit DB. RAF inhibitor resistance is mediated by dimerization of aberrantly spliced BRAF(V600E). *Nature*. 2011;480(7377):387–390. PMCID: PMC3266695.
30. Atefi M, von Euw E, Attar N, **Ng C**, Chu C, Guo D, ... Lo RS, Ribas A. Reversing melanoma cross-resistance to BRAF and MEK inhibitors by co-targeting the AKT/mTOR pathway. *PLoS One*. 2011;6(12):e28973. PMCID: PMC3237573.
31. Niehr F, von Euw E, Attar N, Guo D, Matsunaga D, Sazegar H, **Ng C**, Glaspy JA, Recio JA, Lo RS, Mischel PS, Comin-Anduix B, Ribas A. Combination therapy with vemurafenib and metformin in melanoma cell lines with distinct driver mutations. *Journal of Translational Medicine*. 2011;9:76. PMCID: PMC3152784.
32. Ramey JD, Villareal VA, **Ng C**, Ward SC, Xiong JP, Clubb RT, Bradley KA. Anthrax toxin receptor 1/tumor endothelial marker 8: mutation of conserved inserted domain residues overrides cytosolic control of protective antigen binding. *Biochemistry*. 2010;49(34):7403–7410. PMCID: PMC2942075.

REVIEWS AND COMMENTARY

33. † Issuree PDA, **Ng CP**, Littman DR. Heritable gene regulation in the CD4:CD8 T cell lineage choice. *Frontiers in Immunology*. 2017;8:291. PMCID: PMC5360760. (co-first author)
34. † **Ng CP**, Littman DR. Tcf1 and Lef1 pack their own HDAC. *Nature Immunology*. 2016;17(6):615–616. PMCID: PMC5675014. (invited News & Views)

PRESENTATIONS

Oral Presentations

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| Mar 2027 | “Mitochondrial ATP fuels T cell effector gene expression.” Cancer Biology and Genetics Meet Up, Memorial Sloan Kettering Cancer Center, New York, NY. (scheduled) |
| Sep 2026 | “Metabolic rewiring of T cells to enhance antitumor immunity.” MSK Postdoctoral Research Symposium, Memorial Sloan Kettering Cancer Center, New York, NY. (scheduled) |

- Sep 2026 "Getting T cells fit to fight cancer." Finalist, three-minute talk competition, Office of Postdoctoral Affairs, Memorial Sloan Kettering Cancer Center, New York, NY.
- 2024–2025 Research-in-progress talks, Physician-Scientist Training Program, Department of Pathology and Laboratory Medicine, Weill Cornell Medicine, New York, NY.

Poster Presentations

- Oct 2026 "Mitochondrial ATP promotes T cell differentiation through chromatin accessibility." Pan-NYC Cancer Postdoc Symposium, NYU Langone Health, New York, NY. (accepted)
- Apr 2026 "Mitochondrial ATP production promotes T cell differentiation and function by regulating chromatin accessibility." IMMUNOLOGY2026, American Association of Immunologists Annual Meeting, Boston, MA.
- Nov 2022 "Evaluation of cell-free DNA testing with Guardant360 CDx for identifying clinically actionable mutations in lung cancer patients at Weill Cornell Medicine." Association for Molecular Pathology Annual Meeting, Phoenix, AZ.

TEACHING AND MENTORING

- 2022–present Graduate student co-mentor, Thompson laboratory, Memorial Sloan Kettering Cancer Center — co-mentored a doctoral student through experimental design, controls, and manuscript preparation for his first paper
- 2023–present Research technicians (2), Thompson laboratory, Memorial Sloan Kettering Cancer Center
- 2022–present Histology and morphologic pathology instruction for postdoctoral fellows entering from clinical and basic science backgrounds, Memorial Sloan Kettering Cancer Center
- 2020–2023 Clinical teaching of junior residents across anatomic pathology services, and supervision of MD/PhD students on the autopsy service, Weill Cornell Medicine
- 2020–2023 MD/PhD rotation students, Department of Pathology and Laboratory Medicine, Weill Cornell Medicine
- 2014–2018 Undergraduate research mentor, Littman laboratory, New York University

REFERENCES

Postdoctoral Advisor

Craig B. Thompson, MD

Professor, Cancer Biology and Genetics Program
Memorial Sloan Kettering Cancer Center

Doctoral Advisor

Dan R. Littman, MD/PhD

Helen L. and Martin S. Kimmel Professor of Molecular Immunology
New York University Grossman School of Medicine · Howard Hughes Medical Institute

Physician-Scientist Training Program Co-Director

Matthew B. Greenblatt, MD/PhD

Associate Professor of Pathology and Laboratory Medicine
Weill Cornell Medicine

Scientific Advisory Committee Member

Agnel Sfeir, PhD

PaineWebber Chair in Cancer Genetics
Molecular Biology Program, Memorial Sloan Kettering Cancer Center